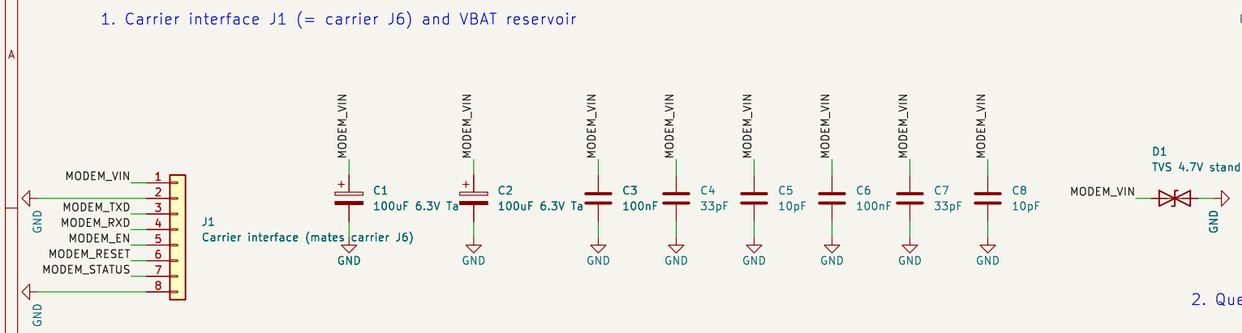
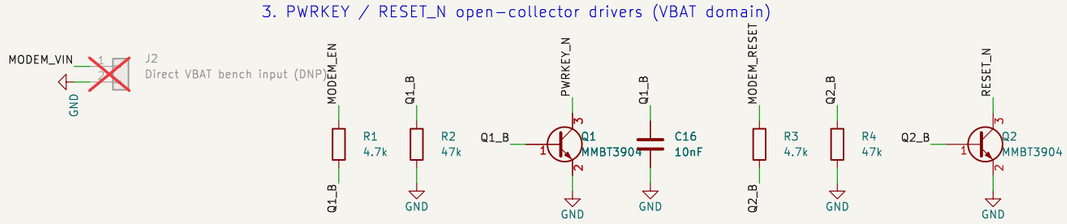
1. Carrier interface J1 (= carrier J6) and VBAT reservoir

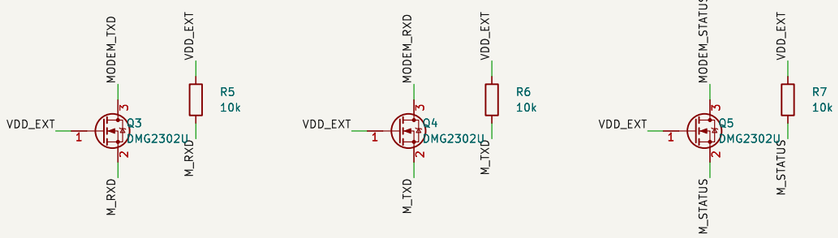


Power: MODEM_VIN (3.8 V nom, 3.3–4.3 V, up to 2 A bursts) comes from carrier J7 through J6 pin 1 (or J2 if fitted). Never from +3V3.

3. PWRKEY / RESET_N open-collector drivers (VBAT domain)

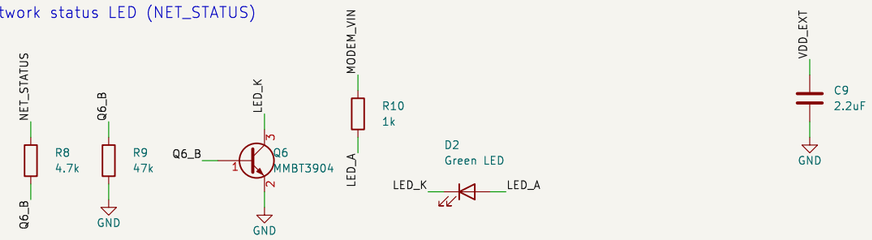


4. 3.3 V <–> 1.8 V level shifting (UART + STATUS)

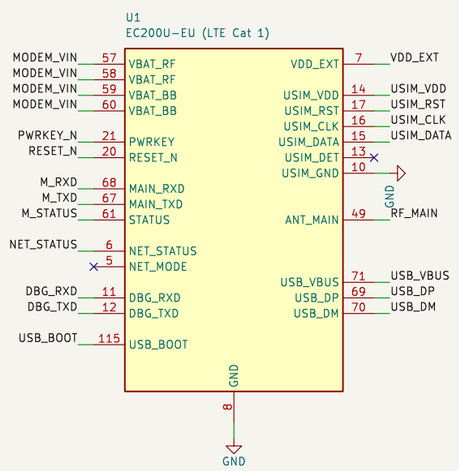


Level shifting: MCU-side pins 3, 4 and 7 have no pull-up on this board (J6 carries no 3V3); firmware enables the ESP32 internal pull-ups on MODEM_RXD and MODEM_STATUS.

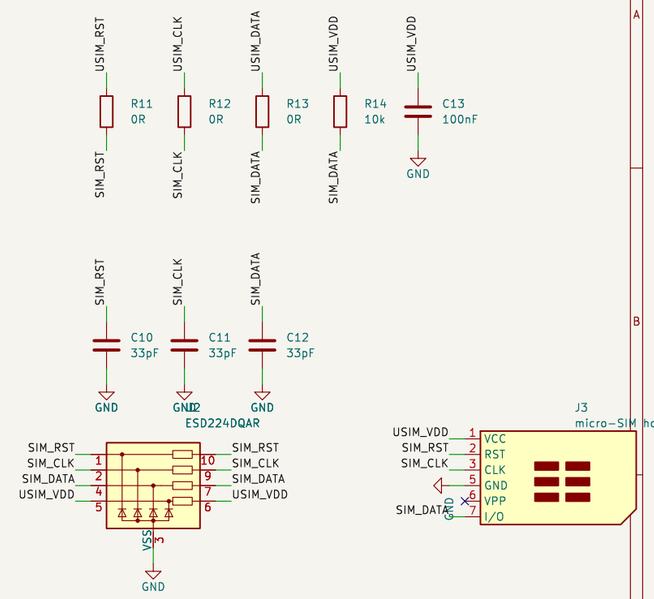
5. Network status LED (NET_STATUS)



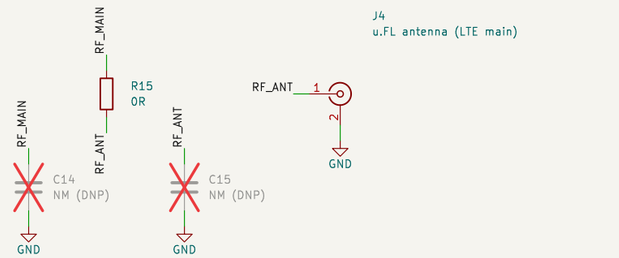
2. Quectel EC200U-EU LTE Cat 1 module (footprint PROVISIONAL)



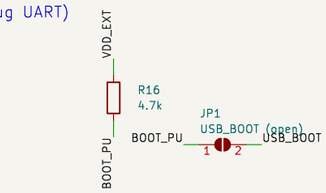
6. micro-SIM: 0R series, 33 pF, pull-up, low-capacitance ESD



7. LTE main antenna: pi network + u.FL (50 ohm)



8. USB download jumper and test pads (1.8 V debug UART)



ENGINEERING DRAFT – NOT FABRICATION READY. Generated from design_modem_v01.py.
EC200U FOOTPRINT IS PROVISIONAL: replace with the Quectel footprint file before layout review.
Bench-only: MODEM_VIN 3.8 V from a current-limited supply. Never on a live gas meter or gas line.
Mates carrier v0.2 J6 (1x08). 3.3 V logic at the header, 1.8 V at the module.

Smart Gas Meter IoT – lab prototype

Sheet: /
File: sgm_modem.kicad_sch

Title: Smart Gas Meter IoT – cellular modem daughterboard v0.1 (EC200U)

Size: A3	Date: 2026-10-26	**Rev: 0.1-draft**
KiCad E.D.A.	kicad-cli 7.0.11+dfsg-1build4	Id: 1/1